Group: Jason Antunes, Gwenn Ashkevron, Wilmer Tobar, Thanapol Shongtuptim, Linya Wang

MGMT 6219 Final Report

Executive Summary

In this report, an analysis covers the Ada Analytics business cycle from August 2025 to January 2026. By tracking month-over-month trends and consultation sessions, clients were strategically segmented based on ROI into four groups: Core Growth Cluster (CGC), Efficiency Focus Group (EFG), Potential Customer Group (PCG), and Marginal Customer Group (MCG). It was found that CGC contains clients with high working hours and rates who maintain a stable current state. EFG contains clients with high maintenance and low hourly rates, requiring improvement in profit margin. PCG contains clients with an extremely high rate but low business volume, making them a key area for future scaling. Lastly, MCG contains clients with both low working hours and rates, therefore having minimal impact on the company's overall revenue.

To support the scaling growth of Ada Analytics, several features were also developed and iterated upon: a What-If Simulation, which introduces a price change percentage slicer to simulate the dynamic impact of pricing adjustments on predicted revenue in real time; Attendance Logic, which implements time-based automatic classification logic (less than 0.05 hrs flagged as a no-show) to identify inefficient behaviours that generate opportunity costs; and Aging Visibility, which enhances monitoring of $9K in revenue at risk from invoices overdue by more than 30 days, guiding targeted collection efforts.

When designing the dashboard, there were two audiences kept in mind. Account managers are the primary audience for this interactive dashboard. Utilisation will assist in identifying high-value clients, monitoring consultation time efficiency, and intervening on client attendance status. Administrators are the secondary audience for this interactive dashboard. Utilisation will allow for review of total revenue metrics on a quarterly basis to assess company growth, manage risk using the outstanding by-aging charts, make strategic decisions, and adjust company resource allocation.

It is recommended that Ada Analytics considers fixing their payment collection process, improving their revenue efficiency per hour, and introduce a KPI monitoring dashboard. Next steps include implementing these recommendations, utilizing the new dashboard, and iteratively testing to find new opportunities for adjustment.

Introduction and Business Context

Throughout the MGMT 6219 Data Analytics Capstone course, a partnership with Riipen and Ana Analytics - a San Francisco-based company that leverages big data and machine learning to create advanced yet easy-to-use and interactive analytical tools, such as client-facing dashboards for a variety of applications - resulted in a culminating deliverable which is the dashboard being discussed in this report.

At the time of commencement, Ada Analytics was in need of a dashboard template for administrators at their consulting agency. They lacked a product that could offer high-level strategic insights to scale and grow the business more effectively. Our objective was to deliver a dashboard

product that could fill this need and provide analytical insights beneficial to the consulting account managers.

Methodology

Our overall method was:

1. Collating the data from separate worksheets to a single dataset
2. Standardizing data formats to facilitate analytical models
3. Cleaning data by removing errors and duplicates
4. Exploring the data with visualizations
5. Applying insights from data exploration to business needs
6. Building the dashboard

1. Collating the Data from Separate Worksheets to a Single Dataset

Our data source is an excel document with 7 worksheets. To begin we reviewed these worksheets and found there were variations on what kind of data they kept. The following table has a breakdown of what columns were part of each worksheet. Highlighted fields show the titles for the columns we decided to use in the final dataset.

1 Outstanding Payments	Date	Client		Start	End	(Mins)	Hourly Rate	Charge		Notes			
2 Nov.27 -	Date	Client		Start	End	Total Time (Mins)	Hourly Rate	Charge		Notes			
3 Oct. 26- Nov. 20	Date	Client		Start	End	Total Time (Mins)	Hourly Rate			Notes			
4 Sept. 22-Oct. 12	Date	Client		Start	End	Total Time (Mins)	Hourly Rate	Charge	Payment	Notes			
5 Payment LinksInvoices	Date	Client		Start	End	(Mins)	Rate			Notes	Link/Invoice Sent	Amount	USD
6 Clients with Issues	Date	Client	Subclient	Start	End	Total Time (Mins)	Hourly Rate			Notes			
7 Free Consults	Date	Client		Start	End	Total Time (Mins)	Hourly Rate			Notes			

Data from each worksheet was migrated to a single excel worksheet with care that the columns aligned properly.

There were rows in most of the worksheets that held sum totals for time and charges. We made note of the values, but deleted these from the dataset since they were aggregations and not original data entries.

2. Standardizing Data Formats to Facilitate Analytical Models

Each column's data was then standardized according to a data dictionary. Columns that do not allow nulls are usable for data modeling.

The Attendance Status was added after step 5: reviewing business needs

Field Name	Data Type	Data format	Nulls OK?	Description	Reason for inclusion
Client	Text	"Client" + capitalized letter of the alphabet	No	Anoymous name given to the consultation client	Primary information
Subclient	Text	Typically Initails or generic description	Yes	When a client orgnaization inculdes one on one consultations	Part of the original data provided
Weekday	Text	Three letter abbrevation of the day of the week	No	formulated from date with = TEXT([Date], "ddd")	Part of the original date provided, but we separated the day of the week for faster anaylsis of weekday and date
Date	Date	dd-mm-yyyy	No	Date the consultation took place	Primary Information
Start	Time	hh:mm	No	Start time of the meeting.	Primary information
End	Time	hh:mm	No	End time of the meeting.	Primary information
Total time	Time	[hh]:mm	No	Duration of the meeting. Formulated with =MOD([End]-[Start], 1)	Provided in the orignal data set
Hourly rate	Currency	$ XX.XX	No	Consultation fee for 60 minutes	Primary information
Charge	Currency	$ XX.XX	No	Formulated from hourly rate and total time. Formulated with =[Total time]*24*[Hourly Rate]	Provided in the orignal data set
Charge notes	Text	Short note	Yes	Notes on discounts, if it has been paid, and note on foreign currency used	Provided in the orignal data set
Invoice Date	Date	dd-mm-yyyy	Yes	Date an invoice was sent to the client	Provided in the orignal data set
Invoice Amount	Currency	$ XX.XX	Yes	Amount provided for the invoice.	Provided in the orignal data set
Notes	Text	Longer note	Yes	Notes from the consultant about what was discussedin the meeting	Provided in the orignal data set
Attendance Status	Text	Either "No-show/Cancelled" or "Showed up"	No	If the Total time measuremnt is less than 0.05 hours, then a meeting is considered No-show. Else it is considered Showed Up.	Allows for measurement of clients that attend meetings with meaningful durations

3. Cleaning Data by Removing Errors and Duplicates

There were five possible miscalculations or errors found in the original dataset as we compared the original data provided to the calculations done by Excel.

Client	Subclient	Weekday	Date	Start	End	Total Time	Time (Mins) - Given	Hourly Rate	Charge - Calculated	Charge - given
Client C		Thu	06-11-2025	13:00	14:00	01:00	73.75	$ 147.50	$ 147.50	
Client C		Thu	06-11-2025	14:00	15:03	01:03	73.75	$ 147.50	$ 154.88	
Client D		Tue	04-11-2025	13:00	13:56	00:56	62.5	$ 225.00	$ 210.00	
Client D		Tue	04-11-2025	14:00	15:29	01:29	62.5	$ 225.00	$ 333.75	

The highlighted data is from our source dataset. But when total time or total charge was calculated through Excel, these discrepancies showed up.

There may be individual reasons as to why these are different from the calculations, but for the purposes of our work, we will be removing them and using the calculated amounts for data analysis.

Removing Outliers

Only one outlier was removed, and it was also from a row that had incomplete data. It was an outlier in the hourly rate column and may have represented a one-time charge, but there were no notes to support it.

Removing Duplicates

18 duplicates were removed. These probably came from multiple meetings being represented in multiple worksheets.

Filling in Missing Values

Filled in 9 missing values in the hourly rate column. Our assumption in filling them in was to use the mode hourly rate for that client. We chose mode instead of average because in this context it made

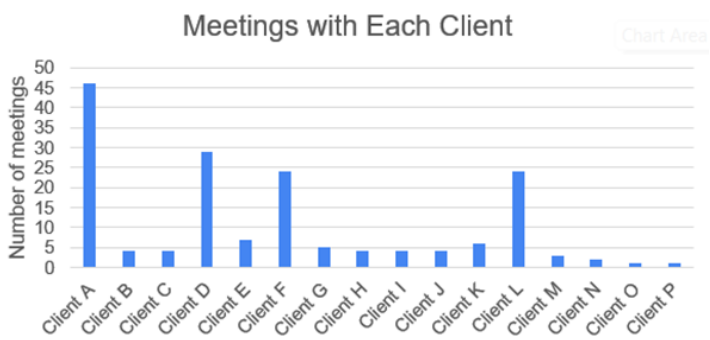
sense that blanks would mean there was no change to the determined hourly rate, and any differences would be noted separately.

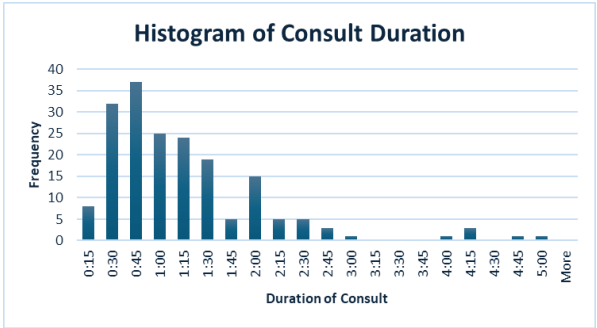
Filled in missing information for meeting start and end times. Again, we went with the mode to copy the most common start time and duration for that client on that particular day of the week.

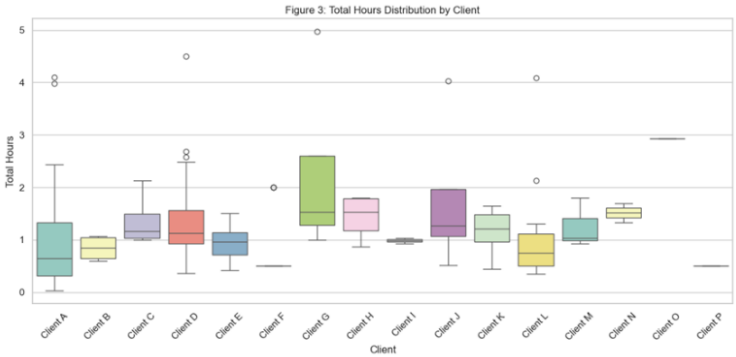
4. Exploring the Data with Visualizations

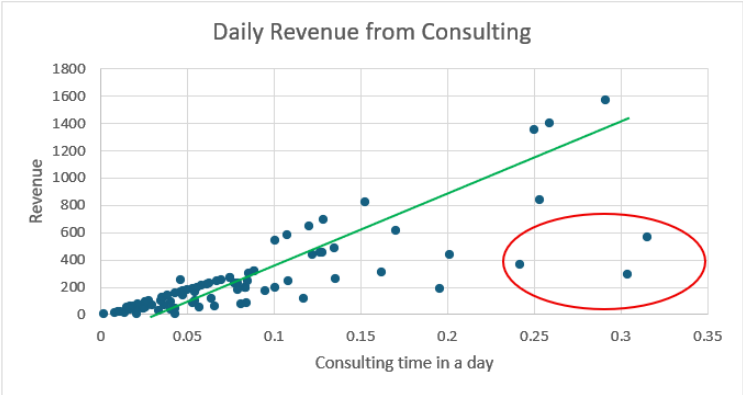
We used both Excel and Python for data visualizations. Histograms, box and whisker, and bar charts let us compare values across time and clients. Scatter plots showed how variables such as Total Time and Charge compared to each other.

Examples below. See Exploratory Data Analysis Report for more detail.









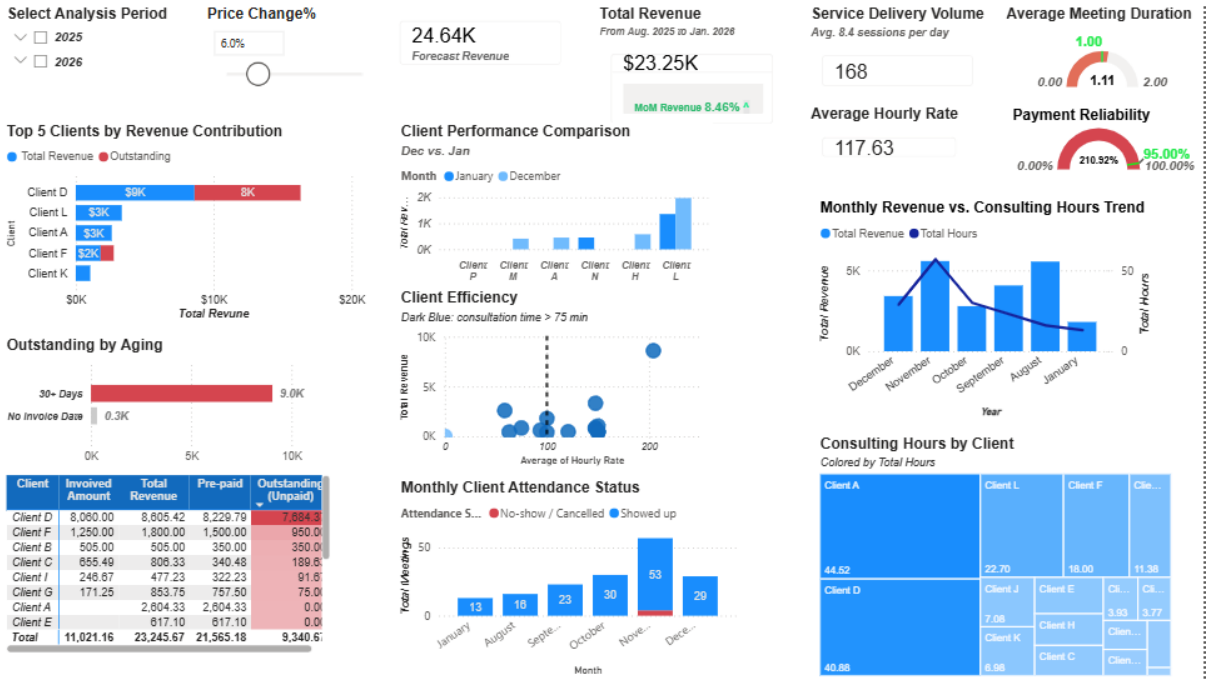
5. Applying Insights from Data Exploration to Business Needs

After reviewing our findings with our client, we decided to focus our findings by looking at how performance impacts revenue and what resources each client requires.

Key Performance Indicators:

- Revenue over time
- Revenue by client
- Total hours in meetings vs revenue
- Hours in meetings by client
- Client attendance
- Unpaid invoice amount by client

6. Building the Dashboard



Client	Invoiced Amount	Total Revenue	Pre-paid	Outstanding (Unpaid)
Client D	8,060.00	8,605.42	8,229.79	7,884.3
Client F	1,250.00	1,800.00	1,500.00	950.0
Client B	505.00	505.00	350.00	350.0
Client C	655.49	806.33	340.48	189.6
Client I	246.67	477.23	322.23	91.6
Client G	171.25	853.75	757.50	75.0
Client A		2,604.33	2,604.33	0.0
Client E		617.10	617.10	0.0
Total	11,021.16	23,245.67	21,565.18	9,340.6

Our dataset is currently light on invoice, payment, and outstanding balance information. However, we believe that including it in the dashboard will allow for valuable business insights as this program is used by the client.

- The top left corner is a key position in a dataset, so we highlighted the ability to filter the data and test different pricing options.
- Below that key position is information on outstanding balances and revenue by client to help visualize any needs on follow-up for billing.
- To the right of the key position are big picture information tiles such as overall revenue, expected time spent in meetings each day, and the average hourly rate for all clients.

- A few meter visualizations help the business see benchmarks and current performance on payment reliability overall for their clients and whether the business is trending towards short or long meetings.
- Towards the middle are some useful stats that show comparisons between clients and for the same clients over time.
- The dashboard is anchored in the bottom right by a map showing which clients the company is investing the most time into.

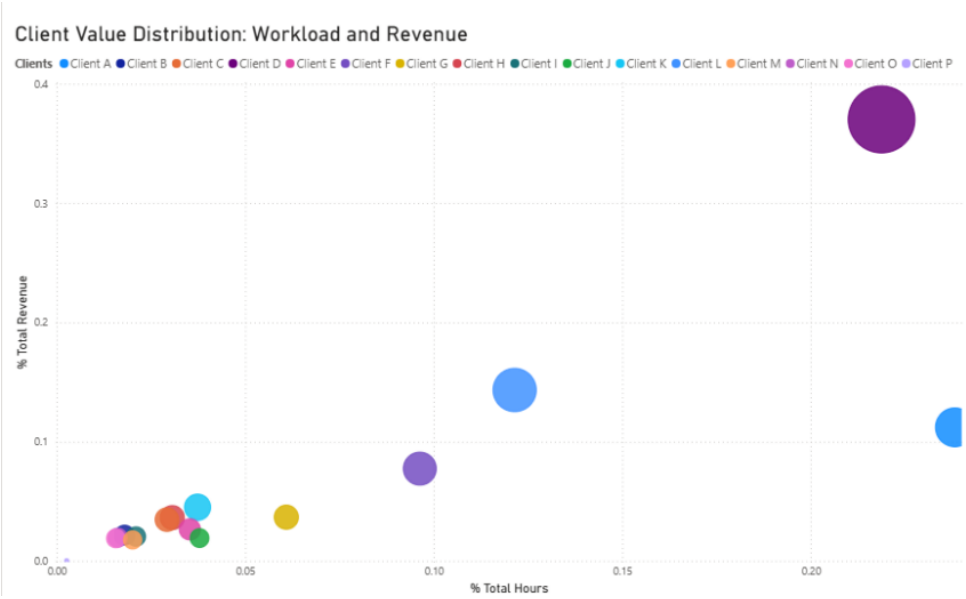
Data Limitations

- This data set is extremely small with less than 200 entries. Even with k-fold cross-validation methods, any machine learning algorithms will be limited in providing accurate predictions.
- Currently the dataset is set up to look at each meeting. However, there are often multiple meetings with the same client on the same day. If we aggregate the data by day, the data set will become even smaller.
- The data spans a date range just shy of 6 months, which limits the accuracy potential with exploration via time-series calculations.
- While we have data on invoices, there may not be enough to provide any useful information on billing cycles or how prompt clients are with their payments.

Findings and Analysis

Client Distribution Summary

The client distribution shows that revenue is concentrated among a few key clients. Client D contributes 37% of total revenue, while Client A has the highest total hours alongside Client D but generates significantly lower revenue due to a much lower and fluctuating hourly rate (ranging from $40 to $100). Clients L and F form a middle segment with moderate hours and revenue, providing a steady contribution. The remaining clients form another segment with lower hours, while most of them have above-average hourly rates (at $117.63) but insufficient hours to make a significant impact.



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The Financial Risk: High Customer Concentration and Payment Collection Crisis

Through in-depth analysis of the financial data, we found that the company is currently in high risk, with the core issues focused on a single customer and failed payment collection process:

Heavy Reliance on Client D:

- Client D is the company's largest revenue source (contributing approximately $9,000), but its outstanding receivables are as high as approximately $8,000. Also, the total revenue from all other major contributing clients is still less than the revenue contributed by client D alone.
- Thus, we find that the company's financial health is heavily reliant on a single customer, and this customer has extremely poor collection ability (outstanding receivables account for over 80% of its total contribution).

Systemic Risks in Payment Collection:

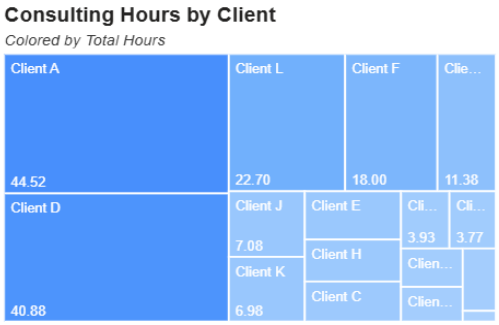
- There is a warning: the 210.92% anomaly displayed on the Payment Reliability dashboard (possibly due to bad debts or calculation errors) further confirms a systemic risk in the collection process.

Refined Finding on Efficiency Gap

By comparing Client D and Client A, we found that high working hours are not always linked to high revenue. This efficiency gap is significantly fluctuating across months:

Benchmark (Client D) vs. Time Sink (Client A):

- Client D generated about $9,000 revenue with moderate working hours. This client can be the efficiency benchmark because its every hour of consultation can be directly converted into high value.
- In the tree map and detailed data tables, Client A consumed the highest amount of consultation time across the company (44.52 hours), but its hourly rate was significantly below average.
- Despite the impressive total time, Client A contributed the only red block (no-show) in the Monthly Client Attendance Status bar chart. This means that the client booked many meetings, but one 30-minute meeting was not attended. Thus, it caused opportunity cost loss.



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Trend Analysis: Seasonal Patterns

Our analysis revealed a negative correlation between seasonal peaks and attendance behavior:

- The Monthly Revenue vs. Consulting Hours Trend shows that November saw a double peak in both hours and revenue, but resource utilization was less than ideal. As shown in the attendance status, increased cancellations/no-shows lead to wasted resources.
- August was the company's most profitable month with fewer working hours. The total consultation time was not the highest in August, but the performance was very stable. It indicates August is an ideal profit model.

Recommendations and Next Steps

1. Fix the Payment Collection Process

- Strengthen payment terms for high-risk clients, as this will help to improve the improve and shorter the payments cycle by defining when a customer should pay its invoice. Payment terms can be defined within 7, 15, or 30 days after the consulting.
- Add a policy for partial upfront deposit (30% to 50%) prior to the consulting.

2. Improve Revenue Efficiency per Hour.

- Adjust the pricing structure, as some clients, like client A, consume the most consulting hours but generate the least revenue.
- Implementing a minimum floor rate aligned with your average hourly rate will be a good base to define a good pricing structure.
- Define client tiers to prioritize high-value scheduling, as creating tiers will help to segment clients by time and revenue generation.
    - For example:
        - Tier 1 → high-rate clients
        - Tier 2 → stable mid-tier clients
        - Tier 3 → low-rate flexible client
- Reduce single client concentration for revenue, as it is the case for client D, which generates 37% of revenue but has 80% of outstanding balances.

3. Introduce a KPI Monitoring Dashboard

This will help with:

- Revenue concentration ratio
- Receivables aging
- Revenue per consulting hour
- Cancellation rate
- Attendance reliability score

- Client profitability index

Resources

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